

DUNDALK INSTITUTE OF TECHNOLOGY
School of Informatics & Creative Arts
Ethical Approval Form for Research Projects

Researcher Name: Harsh Khatri Year: 2026 Course MSC in Data Analytics

Title of project : Predicting Final-Hour SPX Price Movements for Short-Dated Options Trading Using Intraday Market Features

Name of supervisor/s Zohaib Ijaz
(if applicable)

Date 25:03:26

This application is to be completed by the researcher and where appropriate, in conjunction with the project supervisor. The lead researcher/supervisor is responsible for submitting the completed form to the appropriate Research Ethics Committee (details below)

Please note: If your submission is incomplete or unclear, your application will be returned to you, and your project may be delayed.

Section 1

Type of Researcher

Please tick the appropriate box below to indicate the type of researcher you are:

Undergraduate
(proceed to section 2)

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Completed Ethical Approval forms for undergraduate research should be submitted to the relevant Departmental Research Ethics Committee (DREC).

Drec.dcam@dkit.ie – Department of Creative Arts Media and Music

Drec.dcs@dkit.ie – Department of Computing Science and Mathematics

Drec.dvhcc@dkit.ie – Department of Visual and Human Centred Computing

Postgraduate
(proceed to section 3)

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Completed Ethical Approval forms for Postgraduate research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

Staff
(proceed to section 3)

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Completed Ethical Approval forms for Staff research should be submitted directly to the School Research Ethics Committee (SREC).

Srec.ica@dkit.ie

Section 2

Please complete questions 1-4 listed below.

	Human and / or Animal Research	YES	NO
1	Does your research involve human participants other than the following¹? • Research using exclusively secondary sources. • Research using materials legally accessible to the public that have legal protection, e.g., record of court judgements, data archives. • Research using materials that are publicly accessible and where there is no reasonable expectation for privacy, e.g., books, published third party interviews. • Observations of human behaviour in public where (i) those being observed have no reasonable expectation of privacy, (ii) there is no intervention on the part of the researcher nor any interaction between the researcher and those observed, and (iii) individuals are not identifiable in the results. If 'YES', please complete B and C below.		
2	Does your project involve working with animals? – If 'YES', please complete B and C below. – Please note that for ethical consideration: 'Animals' are classed as vertebrate animals including cyclostomes and cephalopods (DIRECTIVE 2010/63/EU)		
3	Does your project involve working with participants from any of the following categories? • Minors (under 18 years of age) • People with learning or communication difficulties • Patients • People in custody • People engaged in illegal activities If 'YES', please complete D below.		
4	Does your project have any possible ethical implications other than those outlined in questions 1, 2 and 3? If 'YES', please complete E below.		

A. I consider that this project has no significant ethical implications² to be brought through the ICA School Ethics Review Process
Please complete Section 4

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¹ If there is any doubt, researchers should contact the Chair of the SREC.

² In determining significant implications, please consider all potential risks attached to this project. If there is any doubt, researchers should contact the Chair of the SREC.

B.

I. Is this study part of a larger project that already has ethical clearance?

YES / NO

If **YES** please answer question B II.

If **NO** please answer question C.

II. If this study is part of a larger project that already has ethical clearance, are you proposing any changes to the operational plan already ethically approved?

YES / NO

If **YES**, please complete *Sections 3 and 4*.

If **NO**, then please provide the project details below and complete *Section 4*.

Title of project with ethical clearance: _____

C. Could this project have ethical implications that should be brought before the appropriate ICA Departmental Ethics Review Committee as it will be carried out with human participants?

YES / NO

If **YES**, please complete *Sections 3 and 4*.

If **NO**, please complete *Section 4*.

D. I consider that this project may have ethical implications that should be brought before the School Research Ethics Committee as it will be carried out with human participants in a “vulnerable” category.

Please complete Sections 3 and 4

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All research carried out with human participants in a vulnerable category must be referred by the Departmental Research Ethics Committee to the School Research Ethics Committee for approval.

E. Could this project have ethical implications, other than those previously outlined, that should be brought before the appropriate ICA Departmental Ethics Review Committee?

YES / NO

If **YES**, please complete *Sections 3 and 4*.

If **NO**, please complete *Section 4*.

Section 3

3.1 Application Form Checklist

Please complete Section 3 and provide additional information as attachments.

My application includes the following documentation:	INCLUDED (mark as YES)	NOT APPLICABLE (mark as N/A)
Recruitment advertisement		NA
Participant Information Leaflet		NA
Participant Informed Consent form		NA
Questionnaire/Survey		NA
Interview/Focus Group Questions		NA
Debriefing material		NA
Evidence of approval to gain access to off-site location		NA
Ethical approval from external organizations. If ethical approval from external organizations is pending give details below		NA
Details		

3.2 Project Details

a) Lay description (Maximum 200 words)

Please outline, in terms that any non-expert would understand, what your research project is about, including what participants will be required to do. Please explain any technical terms or discipline-specific phrases.

In the last hour of trading, financial markets frequently see increased volatility, especially in the S&P 500 index (SPX), where short-dated options are extremely vulnerable to sudden price swings. Although it is challenging to consistently predict these short-term movements, they can present trading opportunities. This dissertation explores the possibility of using intraday market behavior patterns to forecast notable price changes during the last trading hour. Machine learning models will be created to determine whether the market is likely to move significantly upward, downward, or remain relatively stable prior to market close by analyzing historical price and volatility data. Additionally, the study will assess whether these forecasts can enhance decision-making in a streamlined end-of-day options trading approach. The objective is to close the gap between statistical modeling and real-world trading applications.

b) Research objectives (Maximum 150 words)

Please summarise briefly the objectives of the research.

RQ1:

Can intraday market features be used to predict the direction of SPX price movements during the final hour of trading?

RQ2:

To what extent can intraday features predict the magnitude of SPX price movements during the final hour of trading?

RQ3:

Which intraday features (e.g., momentum, volatility, and price positioning) are most important in predicting large end-of-day movements?

RQ4:

Can SPX price movements in the final hour be predicted with sufficient accuracy to be economically meaningful for trading applications?

RQ5:

Can these predictions be used to construct a profitable end-of-day options trading strategy, after accounting for transaction costs and realistic execution constraints?

c) Research location and duration

Location(s)/Population*	Dundalk Institute of Technology
Research start date	2 nd March 2026
Research end date	September 2026
Approximate duration	6 months

* If location/Population other than DkIT campus/population, provide details of the approval to gain access to that location/population as an appendix.

3.3 Participants

		YES	NO	N/A
Do participants fall into any of the following special groups?	Minors (under 18 years of age)			NA
	People with learning or communication difficulties			NA
	Patients			NA
	People in custody			NA
	People engaged in illegal activities (e.g. drug-taking)			NA
Have you given due consideration to the need for satisfactory Garda clearance?				NA

3.4 Sample Details

Approximate number	NA
Where will participants be recruited from?	NA
Inclusion Criteria	NA
Exclusion Criteria	NA
Will participants be remunerated, and if so in what form? NA	

Justification for proposed sample size and for selecting a specific gender, age, or any other group if this is done in your research.

NA

3.5 Risk to Participants

- Please describe any risks to participants that may arise due to the research. Such risks could include physical stress, emotional distress, perceived coercion e.g. lecturer interviewing own students. Detail the measures and considerations you have put in place to minimize these risks
- What will you communicate to participants about any identified risks? Will any information be withheld from them about the research purpose or procedure? If so, please justify this decision.

3.6 Informed Consent

	YES	NO	N/A
Will you obtain active consent for participation?			NA
Will you describe the main experimental procedures to participants in advance?			NA
Will you inform the participants that their participation is voluntary and may be withdrawn at any point?			NA

If the research is observational, will you ask for their consent to being observed?			NA
With questionnaires, will you give participants the option of omitting questions they do not want to answer?			NA
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			NA
Will the data be anonymous?			NA
Will you debrief participants at the end of their participation?			NA
Will your project involve deliberately misleading participants in any way, or will information be withheld? If you answer yes, give details and justification for doing this below.			NA

a) Please outline your approach to ensuring the confidentiality of data (that is, that the data will only be accessible to agreed upon parties and the safeguarding mechanisms you will put in place to achieve this.) You should include details on how and where the data will be stored, and who will have access to it.

Publicly accessible data and saved on DKIT OneDrive with password and MFA with Private access to Supervision Team and Researcher

b) Please outline how long the data will be retained for, if it will be destroyed and how it will be destroyed.

1.Storage : DKIT one Drive

2. Access: Private

3. Communication : Teams and Outlook

4. Digital Platform Usage : Mahara, DKIT one drive

5. Data maintenance: None destroyed after project

Section 4

Researcher I have read and I understand the DkIT Ethics Policy available from:
<https://www.dkit.ie/assets/uploads/documents/Research/Policies/DkIT%20Research%20Ethics%20Policy.pdf>

Signed: Harsh Khatri Print Name: Harsh Khatri Date: 25/03/26
(Researcher)

Supervisor: Applications for Ethical Approval of Undergraduate projects are forwarded to the Departmental Research Ethics Committee for approval or referral to the School Research Ethics Committee. Applications for Ethical Approval of Postgraduate and Staff projects are sent to the School Research Ethics Committee for Approval.

I have read and approved this form & information:

Signed: zohaib Print Name: Dr. Zohaib Ijaz Date: 25/03/26
(Supervisor/Head of Department/ Research Centre Director/ Head of School)

There is an obligation on the researcher and/or supervisor to bring to the attention of the Departmental/School Research Ethics Committee(s): (a) Any issues with ethical implications not clearly covered by this form (b) Any ethical issues which may arise during the carrying out of the research; (c) Any ethically significant change made to the project after approval.

Section 5 (For office use only)

STATEMENT OF ETHICAL APPROVAL (FOR UNDERGRADUATE PROJECTS ONLY)

This project has been considered using agreed department procedures and is now:

Approved:

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Referred to the School Ethics Committee:

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Signed: _____ Print Name: _____ Date: _____
(Chair of Departmental Research Ethics Committee/Head of Department)

STATEMENT OF ETHICAL APPROVAL

This project has been considered using agreed School procedures and is now:

Approved:

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Rejected (further information sought):

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Chair of School Research Ethics Committee

This project has been considered by the Ethics Committee and ethical approval is granted.

Signed: _____ Print Name: _____ Date: _____
Chair of School Research Ethics Committee